

Management Information Systems

07. Information Gathering

Guo, Xunhua

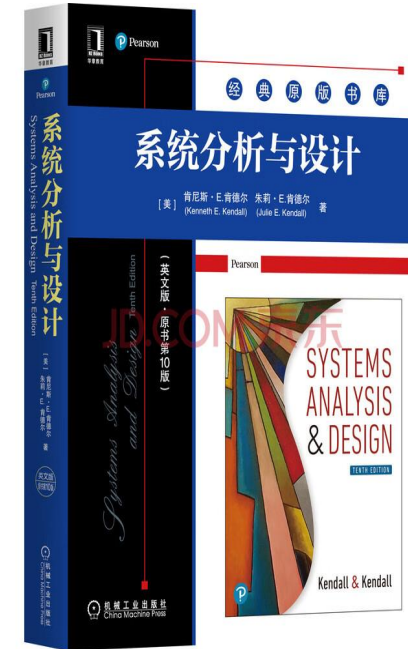
guoxh@tsinghua.edu.cn



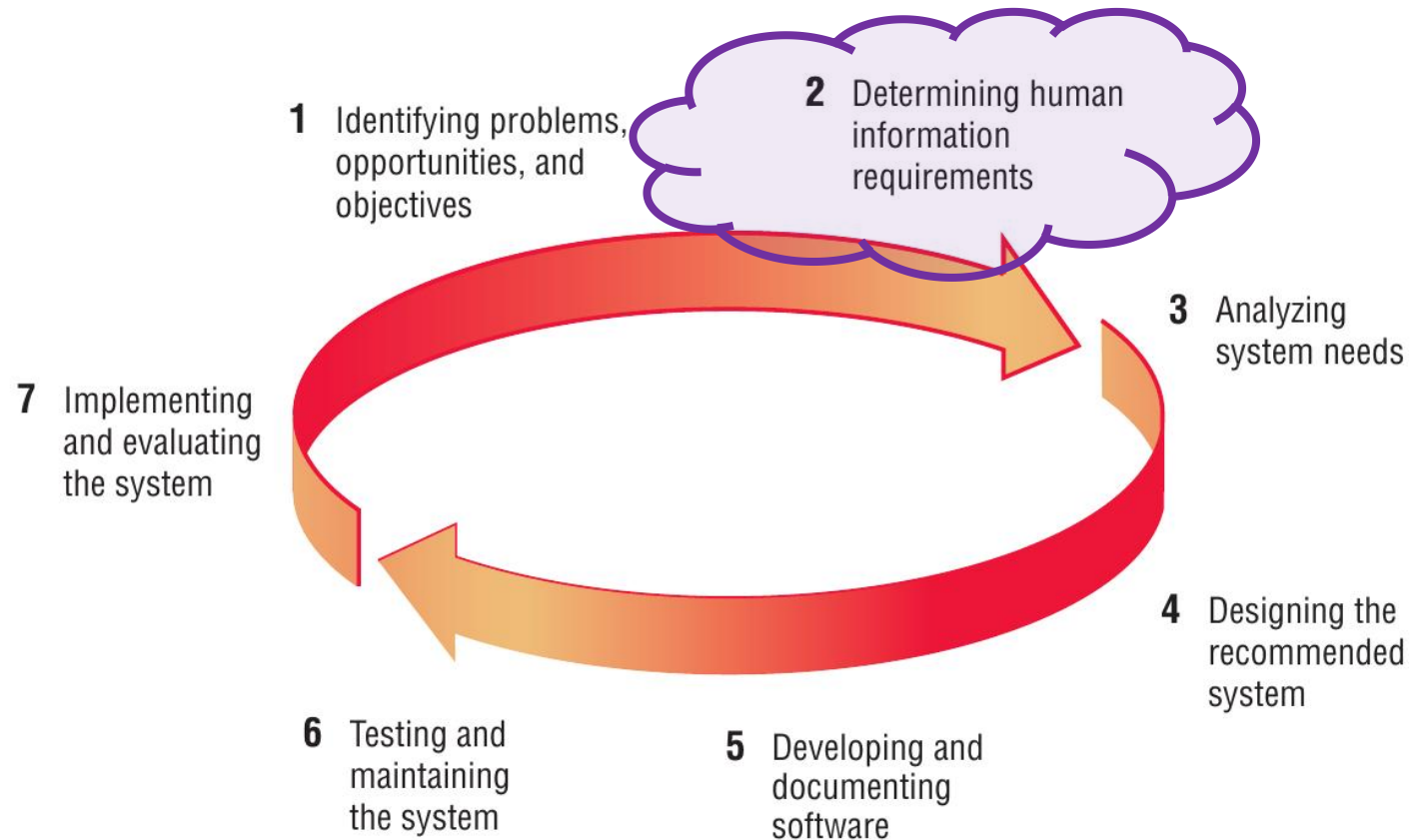
清华经管学院
Tsinghua SEM

Textbook

- Kendall, K. E. and Kendall, J. E. Systems Analysis and Design, (Global Edition, 11th Edition), Pearson Education Inc. (2024).
 - E-book available at:
<https://ereserves.lib.tsinghua.edu.cn/bookDetail/092404F52B635700E065000000000001>



Information Gathering



Information Gathering

- Interactive Methods (Textbook: Chapter 4)
- Unobtrusive Methods (Textbook: Chapter 5)
- AI-based Development Tools



Interactive Methods



Learning Objectives

- 4.1 Recognize the value of interactive methods for information gathering
- 4.2 Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users
- 4.3 Understand the purpose of stories and why they are useful in systems analysis
- 4.4 Understand the concept of JAD and when to use it
- 4.5 Write effective questions to survey users about their work
- 4.6 Design and administer effective questionnaires



Interactive Methods to Elicit Human Information Requirements

- Interviewing
- Joint Application Design (JAD)
- Questionnaires



Interviewing

- Interviewing is an important method for collecting data on human and system information requirements
- Interviews reveal information about:
 - Interviewee opinions
 - Interviewee feelings
 - Goals
 - Key HCI (Human-Computer Interaction, 人机交互) concerns



Interview Preparation

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure



Question Types

Open-Ended Interview Questions

- What's your opinion of the current state of business-to-business ecommerce in your firm?
- What are the critical objectives of your department?
- Once the data are submitted via the Web site, how are they processed?
- Describe the monitoring process that is available online.
- What are some of the common data entry errors made in this department?
- What are the biggest frustrations you've experienced during the transition to ecommerce?

Closed Interview Questions

- How many times a week is the project repository updated?
- On average, how many calls does the call center receive monthly?
- Which of the following sources of information is most valuable to you?
 - Completed customer complaint forms
 - Email complaints from consumers who visit the Web site
 - Face-to-face interaction with customers
 - Returned merchandise
- List your top two priorities for improving the technology infrastructure.
- Who receives this input?

Bipolar Interview Questions

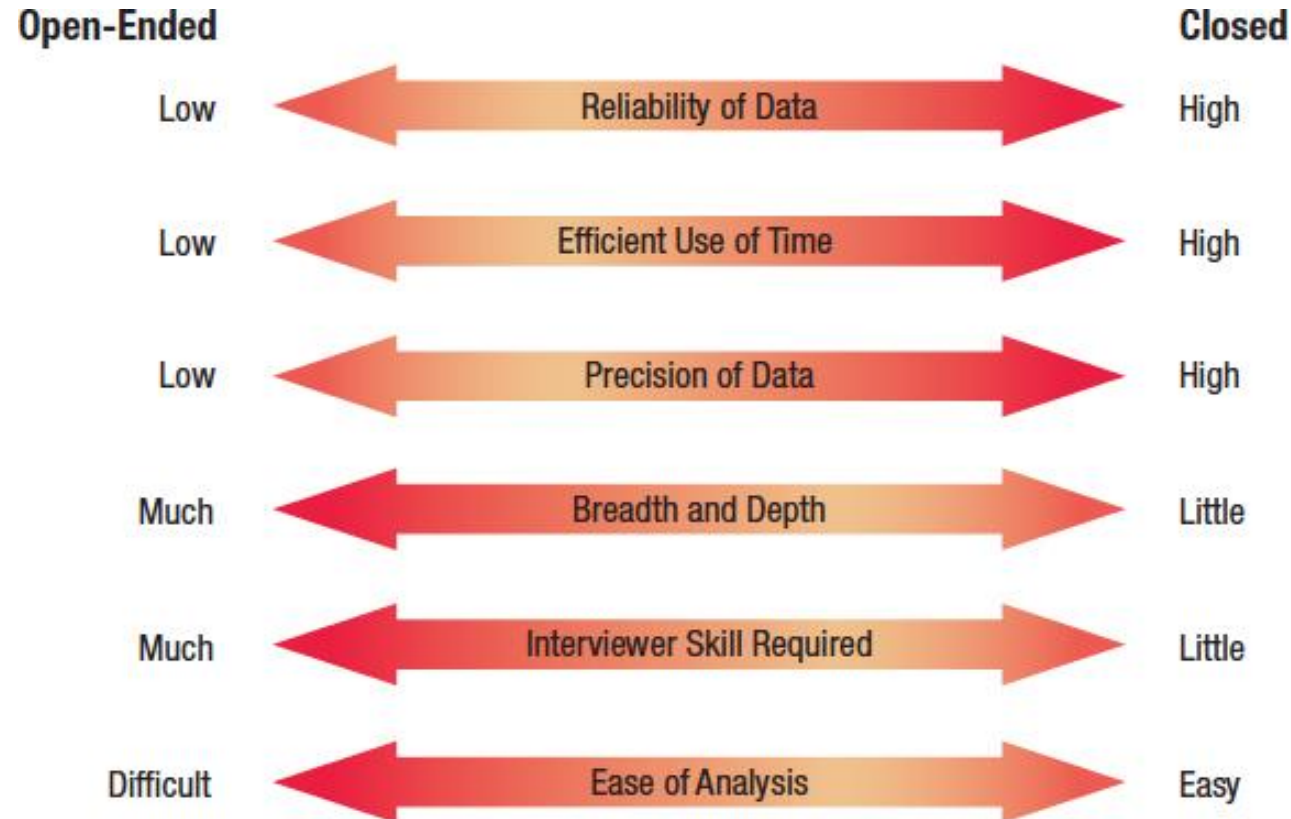
- Do you use the Web to provide information to vendors?
- Do you agree or disagree that ecommerce on the Web lacks security?
- Do you want to receive a printout of your account status every month?
- Does your Web site maintain a FAQ page for employees with payroll questions?
- Is this form complete?

Probes

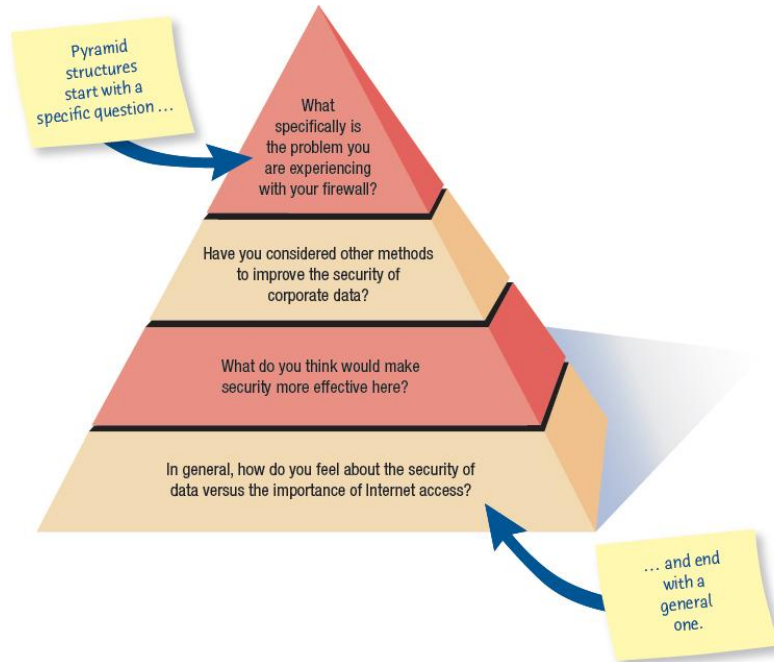
- Why?
- Give an example of how ecommerce has been integrated into your business processes.
- Please give an illustration of the security problems you are experiencing with your online bill payment system.
- You mentioned both an intranet and an extranet solution. Please give an example of how you think each differs.
- What makes you feel that way?
- Tell me step by step what happens after a customer clicks the "Submit" button on the Web registration form.



Open-Ended vs. Closed Questions

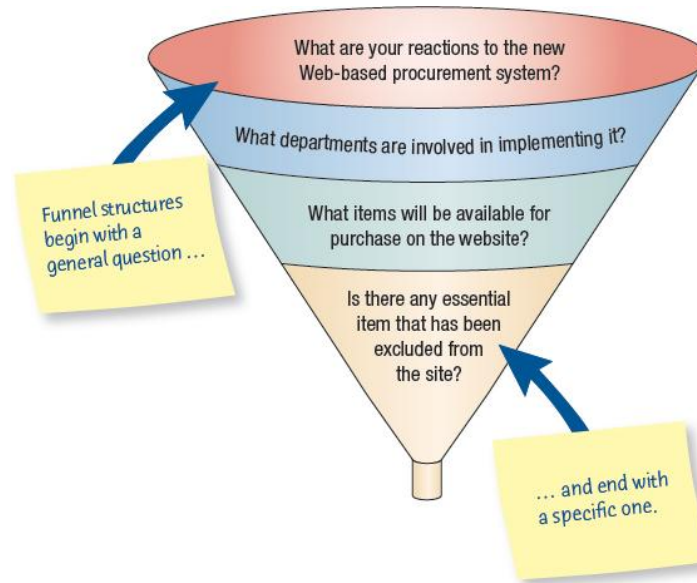


Arranging Questions



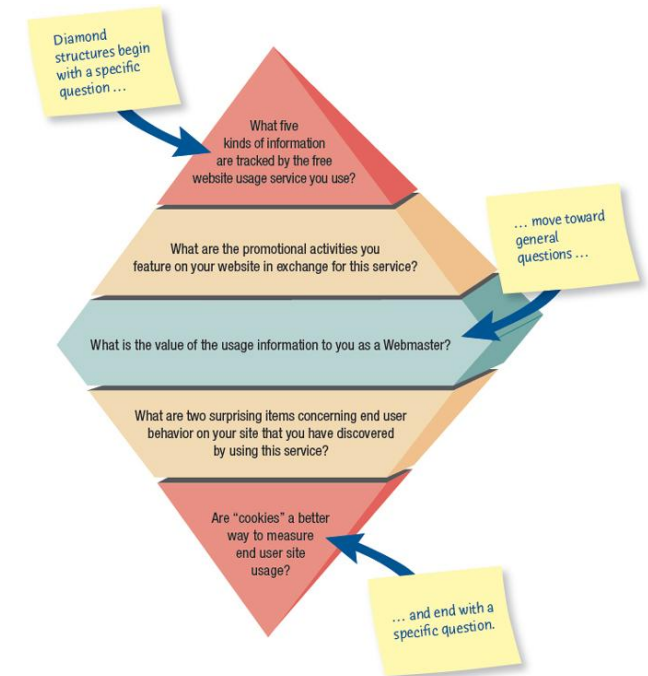
Pyramid Structure (▲): Closed → Open

- useful if interviewees need to be warmed up or seem reluctant to address the topic



Funnel Structure (▼): Open → Closed

useful when the interviewee feels emotionally about the topic



Funnel Structure (▼): Open → Closed

combines the strength of both, takes longer



Closing the Interview

- Always ask “Is there anything else that you would like to add?”
- Summarize and provide feedback on your impressions
- Ask whom you should talk with next
- Set up any future appointments
- Thank them for their time and shake hands.



Interview Preparation: Using Catalyst (启航杯创赛) as an Example

- Reading background material
 - 组织结构图: 组委会架构 (赵主任→莎莎→评委团队)
 - 现有文档: **Excel**打分模板、错误报告、微信聊天记录
 - 系统截图: 手工重命名**PDF**的文件夹、评委抱怨手机打不开**Excel**的截图
 - 对话脚本: 莎莎与赵主任、刘教授的冲突对话
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure



Interview Preparation: Using Catalyst (启航杯创赛) as an Example

- Reading background material
- Establishing interview objectives
 - 目标1：获取“BP匿名化处理”流程细节
 - 目标2：识别“手工Excel汇总”风险点
 - 目标3：了解“评委移动端需求”
 - 目标4：挖掘“版本控制混乱”的根源
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure



Interview Preparation: Using Catalyst (启航杯创赛) as an Example

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
 - 莎莎 (操作层): 详细操作步骤、具体痛点
 - 赵主任 (管理层): 业务目标、风险控制
 - 刘教授 (外部用户): 用户体验、界面需求
- Preparing the interviewee
- Deciding on question types and structure



Interview Preparation: Using Catalyst (启航杯创赛) as an Example

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
 - 例：给莎莎的邀请：
 - “我们正在设计新的'启航杯评分管理系统', 目标是**消除手工错误、提高您的工作效率**。
 - 明天下午3点的访谈，想请您：**1. 描述当前处理BP邮件、重命名PDF、分发打包的完整流程；2. 指出流程中最容易出错、最耗时的环节；3. 分享您对理想系统的设想（哪些操作希望自动化？）**
 - 您的输入将转化为**系统功能设计**，帮助我们构建真正解决您痛点的工具。
 - 访谈约**45分钟**，我会记录并整理成需求文档供您确认。”
- Deciding on question types and structure



Interview Preparation: Using Catalyst (启航杯创赛) as an Example

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure
 - 问题举例（针对目标1）
 - 开放式：“请描述您从收到BP邮件到完成匿名化打包的完整过程。”
 - 封闭式：“匿名化时需要隐藏哪些信息？(A)学生姓名 (B)导师姓名 (C)学校名称 (D)全部”
 - 追问：“500个文件总共处理多久？最耗时的步骤是什么？”
 - 问题结构：采用**钻石结构**（封闭→开放→封闭）



From Dialogue to Requirements - Example 2

- 访谈记录:
 - “每天500封邮件。我逐个下载PDF，重命名为‘项目001’、‘项目002’以隐藏姓名...”

方面	发现	系统需求
痛点	处理流程低效	提升流程效率
功能需求	自动PDF匿名化	UC-01 自动匿名化BP
业务规则	隐藏学生/导师姓名	BR-01 移除可识别信息
质量需求	每日500+ PDF	NF-01 批量处理500+文件



From Dialogue to Requirements - Example 2

- 访谈记录:
 - “去年的错误: Excel公式在复制粘贴时错位...”

方面	发现	系统需求
痛点	手动传输错误	自动传输评分数据
功能需求	自动汇总 + 验证	UC-03 自动汇总评分
业务规则	校验汇总信息	BR-02 交叉验证并告警
质量需求	零手动复制粘贴	NF-03 无手动数据传输



Interview Report

访谈报告：启航杯 - 莎莎（赛事统筹助理），日期：202X-XX-XX，采访者：XXX，目标：评分流程需求获取

- 1. 关键引语（用户原话）
 - “每天500封邮件，我逐个下载PDF，重命名...”
 - “去年：复制粘贴错位错误...”
 - “刘教授：压缩包太大，手机打不开...”
- 2. 需求提取
- 2.1 功能需求：
 - FR-01：自动匿名化上传的BP PDF文件
 - FR-02：移动端网页打分（无需下载附件）
 - FR-03：自动汇总评分（无手动复制）
- 2.2 业务规则：
 - BR-01：评分范围仅为1-10整数
 - BR-02：追踪BP版本，通知评委更新
 - BR-03：评委提交后应收到确认
- 2.3 质量需求：
 - NF-01：移动端加载 < 3秒（弱网环境）
 - NF-02：支持500+并发上传处理
 - NF-03：无手动数据传输
- 3. 关联用例
 - UC-01：匿名化BP
 - UC-02：移动端打分
 - UC-03：自动汇总评分
- 4. 待跟进问题
 - 学校信息也需要匿名化吗？
 - 是否允许评委保存草稿分数？



Joint Application Design (JAD)

- Joint Application Design (JAD, 联合应用设计)
 - can replace a series of interviews with the user community
 - allows the analyst to accomplish requirements analysis the users in a group setting
- Benefits of JAD
 - Time is saved, compared with traditional interviewing
 - Rapid development of systems
 - Improved user ownership of the system
 - Creative idea production is improved
- Drawbacks
 - JAD requires a large block of time to be available for all session participants
 - If preparation or the follow-up report is incomplete, the session may not be successful
 - The organizational skills and culture may not be conducive to a JAD session



Questionnaires

- Why Questionnaires
 - Scale: Reach many stakeholders efficiently
 - Quantify: Transform opinions into comparable data
 - Prioritize: Rank features with statistical confidence
 - Segment: Identify role-based needs
- Key Design Focus
 - Question types
 - Open-ended → Explore new ideas, qualitative insight
 - Closed/scale → Measure attitudes, benchmark satisfaction
 - Ranking → Force trade-off decisions
 - Avoid bias: careful sampling, neutral wording, balanced scales, random order
 - Ensure reliability & validity: Questions must measure what they intend
- Bottom Line
 - Questionnaires provide the quantitative backbone for evidence-based system design
 - complementing interviews/JAD with scalable data.



Unobtrusive Methods



Learning Objectives

- 5.1 Recognize the value of unobtrusive methods for information gathering
- 5.2 Understand the concept of sampling for human information requirements analysis
- 5.3 Construct useful samples of people, documents, and events for determining human information requirements
- 5.4 Interpret managers' and customers' messages, interviews, and communications using text analytics
- 5.5 Create an analyst's playscript to observe decision-maker activities
- 5.6 Apply the STROBE technique to observe and interpret the decision maker's environment



Unobtrusive Methods (无干扰方法)

- What are Unobtrusive Methods?
 - Techniques for gathering information **without direct interaction** with stakeholders
 - **Examine existing artifacts** (documents, emails, systems)
 - **Observe behaviors and environments** (physical workspace, digital traces)
 - **Analyze textual data** (feedback, communications)
- Why use them?
 - Reveal actual vs. stated processes (what people do vs. what they say they do)
 - Identify hidden patterns (data flow bottlenecks, communication gaps)
 - Complement interviews/JAD with objective evidence
 - Scale analysis across large datasets (e.g., 500+ BP submissions)



Major Topics

- Sampling
- Quantitative document analysis
- Qualitative document analysis
- Text analytics
- Observation
- STROBE (**STR**uctured **OB**servation of the **E**nvironment, 结构化环境观察)



Sampling

- Key Concepts
 - Population: All items of interest (e.g., all 500 BP submissions)
 - Sample: A representative subset for detailed analysis
 - Sampling error: Difference between sample and population
- Sampling Methods (Catalyst Examples)

Method	Description	Catalyst Application
Simple Random	Every BP has equal chance	Randomly select 50 BPs for format-consistency check
Stratified	Divide population into subgroups (strata)	Sample separately from: Tech-focused BPs (200), Social-impact BPs (150), Traditional-business BPs (150)
Systematic	Select every k-th item	Review every 10th BP upload to detect naming-pattern violations
Purposive	Hand-pick informative cases	Select 5 BPs that received the widest score variance (controversial projects)



Quantitative document analysis

- Purpose: Extract numerical patterns from existing records
- Example: Catalyst Data Sources
 - Excel scoring logs: Formula errors, completion times, score distributions
 - Email metadata: Response times, thread lengths, attachment sizes
 - System logs: Upload timestamps, file-type distribution, access patterns
 - Dataset: Last year's 50 judge Excel files
- Quantitative findings:
 - 1.35% of files had formula-reference errors (misaligned rows)
 - Average completion time: 4.2 days (range: 1~21 days)
 - 22% of judges submitted scores after deadline
 - 8MB average file size → mobile-loading issues
- Deliverables:
 - Performance baseline for new system (e.g., "reduce submission delay to <2 days")
 - Error-rate metrics to justify automation (e.g., "eliminate 35% formula errors")
 - Data-volume requirements (e.g., "support 8MB files on mobile")



Qualitative Document Analysis

- Purpose: Understand context, meaning, and narratives in documents
- Example: Catalyst Data Sources
 - Email threads: Tone, frustration cues, repeated complaints
 - Meeting minutes: Decision-making process, unresolved conflicts
 - Error reports: Root-cause descriptions, blame assignment
- Qualitative findings:
 - Emotional tone: Frustration ("完全无法使用"), urgency ("3小时内"), threat ("可能无法继续担任评委")
 - Recurring theme: Mobile-incompatibility mentioned 3 times across 6 emails
 - Blame pattern: 刘教授指责组委会，莎莎道歉但提供不了解决办法
 - Power dynamic: 刘教授向赵主任投诉
- Deliverables
 - **User-pain-point narratives** for usability requirements
 - **Communication-flow diagrams** showing escalation paths
 - **Stakeholder-conflict maps** to inform change-management strategy



Text Analytics: Mining Unstructured Feedback

- Purpose: Extract themes, sentiments, and concepts from text data
- Example: Catalyst Text Sources
 - Open-ended survey responses (from our questionnaire)
 - Judge comments in scoring sheets
 - Email bodies (excluding attachments)
 - Chat-log excerpts (WeChat complaints)
- Example Analysis (Catalyst Judge Comments)
 - Text corpus: 200 judge comments from last year's scoring sheets
 - Top concepts: "mobile", "Excel", "format", "time", "confusing"
 - Sentiment: 65% negative, 20% neutral, 15% positive
 - Co-occurrence: "mobile" + "Excel" + "problem" (high frequency)
 - Emerging theme: "version confusion" mentioned by 40 judges
- Deliverables
 - **Priority-requirement list** based on concept frequency
 - **Sentiment-weighted features** (address most negative topics first)
 - **Taxonomy of user complaints** for error-handling design



Observation – The Analyst's Playscript

- Purpose: Record decision-makers' actions in real-time using action verbs
- Method: Create a playscript table with columns:
 - Decision Maker (who)
 - Action (what they do-verb)
 - Information-related Activity (what data they use/produce)
- Insights
 - **Bottleneck:** Manual renaming (5-10 seconds per file $\times$ 500 = 1+ hour)
 - **Risk point:** Large ZIP creation (judges complain about download)
 - **Data-flow gap:** No automatic linkage between email receipt and tracking sheet

Catalyst Example: Observing 莎莎's Morning Routine

Decision Maker	Action	Information-related Activity
莎莎	Opens email client	Receives 87 new BP-submission emails
莎莎	Downloads PDF attachments	Stores files in "Raw_BP_2026" folder
莎莎	Renames files (Project001.pdf)	Transforms identifiable names to anonymous codes
莎莎	Creates ZIP archive	Packages 50 BPs for Judge Liu
莎莎	Sends email with attachment	Transmits 85MB ZIP (mobile-unfriendly)
莎莎	Updates Excel tracker	Records transmission time, judge name



STROBE(Structured Observation of the Environment, 结构化环境观察)

- Purpose: Infer decision-maker characteristics from physical/digital workspace
- Seven Observable Elements (adapted for digital/remote work):
 - 1.Workspace location & layout (office/remote, desk arrangement)
 - 2.Digital-tool placement (multiple screens, device sprawl)
 - 3.Stationary equipment (storage systems, filing habits)
 - 4.Props & personal artifacts (sticky notes, printed lists, personal devices)
 - 5.External information sources (open browser tabs, reference materials)
 - 6.Lighting & color (screen settings, interface themes)
 - 7.Clothing & appearance (professional/casual, web-cam visibility)



Catalyst STROBE Analysis



STROBE Element	Observation	Inference
Digital-tool placement	3 monitors: email, folder explorer, Excel	Multitasking overload – constantly switching contexts
Props	Phone charging, multiple coffee cups	Interruptions – calls/drinks break flow
External information sources	Browser tabs: Email, WeChat, university portal	Fragmented attention – juggling multiple channels
Storage habits	Desktop cluttered with "temp" folders	Ad-hoc organization – no systematic filing
Interface themes	Dark-mode IDE, light-mode Excel	Visual-cognitive switching – eye-strain risk

• STROBE Symbols for Evaluation

- ✓ Narrative confirmed (莎莎确实被邮件淹没); X Narrative reversed (莎莎并非低效, 而是系统低效); ◇ Cue to look further (为何需要3个显示器?); ○ Narrative supplemented (莎莎的咖啡消耗量暗示长时间加班)

Applying STROBE to Catalyst (Example Table)

- Anecdotal (轶事) List with STROBE Symbols

Narrative from Interviews	STROBE Element	Observation	Symbol	Interpretation
"莎莎 is overwhelmed by emails"	Digital-tool placement	3 monitors, 100+ unread emails	✓	Confirmed – visual evidence of overload
"Judges can't use Excel on mobile"	Props	Screenshot: iPhone showing garbled Excel	✓	Confirmed – direct evidence of problem
"We have strict version control"	Storage habits	Desktop: "BP_v1", "BP_v2_final", "BP_new"	X	Reversed – naming inconsistency shows chaos
"Communication is efficient"	External info sources	WeChat: 200+ unread group messages	◇	Look further – apparent inefficiency
"Scoring is objective & fair"	Interface themes	Excel: manual formula editing visible	□	Modifies narrative – human intervention introduces bias

Synthesizing Unobtrusive Findings: Catalyst Example

- Data Triangulation: Combine multiple unobtrusive sources
 - 1.Document analysis: Excel logs show 35% formula errors
 - 2.Text analytics: Judge comments highlight "Excel-mobile" problem
 - 3.Observation: 莎莎 spends 1+ hour renaming files
 - 4.STROBE: Cluttered desktop indicates ad-hoc versioning
- Deliverables Generated

发现	需求	优先级
手动重命名每天消耗>1小时	自动匿名化 (UC-01)	高 (High)
35%的Excel公式错误	自动汇总与验证 (UC-03 + BR-02)	关键 (Critical)
移动端-Excel不兼容	移动优先网页界面 (UC-02 + NF-01)	高 (High)
版本命名混乱	集中式版本控制 (数据模型需求)	中 (Medium)
3台显示器可见的邮件过载	集成通知中心 (系统集成需求)	中 (Medium)

High/Critical: Issues with **multiple confirming sources**; **Medium:** Issues with **strong qualitative evidence**; **Low:** Issues with **single-source support**



Summary: Unobtrusive Methods

- Unobtrusive $\neq$ passive
 - Requires systematic sampling, analysis, and synthesis
- Reveals hidden truths
 - What people actually do vs. what they say they do
- Scales analysis
 - Handle 500 BPs, 50 judges, 1000+ emails
- Provides objective evidence
 - Numbers, screenshots, logs vs. subjective opinions
- Complements interactive methods
 - Interviews reveal "why"; unobtrusive shows "what"



Information Requirement Analysis

- Outcome: Requirements-Specification Document (需求说明书)
 - Project overview (business context, problem statement, objectives)
 - Stakeholder analysis (roles, goals, pain points, influence)
 - Functional requirements (prioritized use-case list, detailed use-case descriptions)
 - Non-functional requirements (performance, usability, reliability, security)
 - Business rules (domain-specific constraints and policies)
 - Data requirements (key entities, attributes, relationship sketch)
 - Assumptions & constraints (technical, business, resource limitations)



AI-based Development Tools



AI-based Development Tools

- AI Code Assistants/IDE Plugins
- Autonomous Coding Agents/CLI Tools
- AI Agent Orchestration/Workflow Frameworks



AI Code Assistants/IDE Plugins

- Real-time, editor-integrated tools for inline code completion, suggestions, and lightweight code generation.
- **GitHub Copilot:** Classic AI pair programmer
 - provides real-time code suggestions, function completions, and snippets directly in IDEs (VS Code, JetBrains, etc.)
 - based on context from your code and open-source repos.
- **Cursor:** AI-first code editor
 - with built-in chat, code refactoring, and context-aware suggestions; designed for tight AI integration throughout the coding workflow.



Autonomous Coding Agents/CLI Tools

- Standalone, multi-step agents that operate via command line—can explore files, refactor codebases, run tests, and complete complex projects end-to-end.
- **Claude Code:** Anthropic's CLI-based coding agent
 - excels at iterative, multi-file tasks (refactoring, feature building, PR reviews) with strong reasoning and safety controls.
- **OpenCode:** Open-source CLI coding agent
 - similar to Codex/Claude Code, designed for autonomous development workflows (requires PTY for interactive use).



AI Agent Orchestration/Workflow Frameworks

- Platforms for managing AI agent sessions, skills, tools, and collaboration—provides the infrastructure to build and run custom agent workflows.
- **OpenClaw:**
 - Manages sessions, sub-agents, skills (like the coding-agent skill), tools, and memory; enables multi-agent collaboration and durable workflow orchestration.
- **Hermes:**
 - Repetitive, structured tasks with consistent users/teams (e.g., regular code reviews, document processing, research workflows) where long-term improvement adds value.



Tasks

- Individual tasks
 - Reading: Textbook, Chapters 4~5
 - Answer the review questions (on YuKeTang)
- Group tasks (all groups)
 - Assignment #7: Information Gathering for Requirements Elicitation
 - Submit your report on WebLearning (网络学堂)
- Class presentations (April 27)
 - Group 3 (王廷轩组, 智云科技) and Group 4 (蓝天翔组, 卢米纳电子)
 - Present your reports
 - Time: 10~20 minutes

